Bluetooth Controlled Fire Fighting Robot Using Arduino Nano

Janmejoy Chakraborty

Department of Electronics and Communication Engineering
Dr. B. C. Roy Engineering College, Durgapur, India
janmejoychakraborty1020@gmail.com

Abstract

Robotics has gained significant importance in safety and automation applications. This project presents the design and development of a **Bluetooth Controlled Fire Fighting Robot using Arduino Nano**. The robot can be controlled wirelessly through a smartphone using an **HC-05 Bluetooth module**.

The system uses **three flame sensors to detect fire** and activates a **water pump through a relay module** to extinguish it. A **servo motor controls the direction of the water spray**, while a **buzzer and LED indicator provide alerts** when fire is detected.

The robot is built on a **two-layer wooden chassis**, where the water tank is separated from electronic components to improve safety and organization. The system demonstrates an effective low-cost firefighting prototype suitable for robotics education and automation demonstrations.

Keywords

Arduino Nano, Bluetooth Control, HC-05, Fire Fighting Robot, Flame Sensor, L298N Motor Driver

I. Introduction

Fire accidents can cause serious damage to property and human life. Early detection and rapid response are critical to minimize losses. Robotics technology can play an important role in assisting firefighting operations, especially in hazardous environments where human access may be risky.

This project demonstrates a **Bluetooth controlled firefighting robot** that can detect fire and spray water automatically. The robot is controlled via a smartphone and uses flame sensors to detect fire. The system integrates **Arduino Nano, Bluetooth communication, sensors, and motor control**, providing a practical example of embedded systems and robotics.

II. System Overview

The system consists of the following main components:

• Smartphone for sending control commands
• HC-05 Bluetooth module for wireless communication
• Arduino Nano as the central controller
• L298N motor driver for motor control
• Flame sensors for fire detection
• Relay module and water pump for extinguishing fire
• Servo motor to direct the water nozzle

The smartphone sends commands through Bluetooth to the HC-05 module. Arduino processes these commands and controls the motors accordingly. When fire is detected, the robot activates the water pump and aims the nozzle toward the flame.

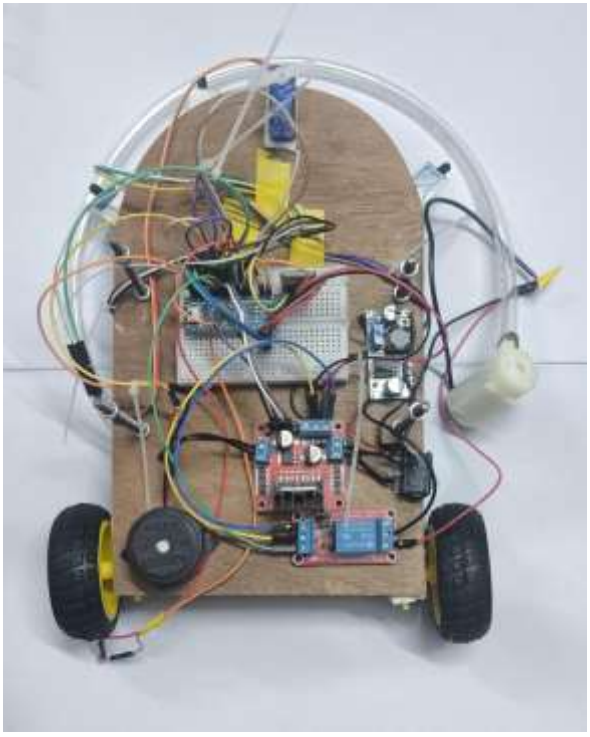

Fig. 1. *The Bluetooth controlled RC fire car front*

III. Hardware Components

- Arduino Nano (ATmega328P microcontroller)
- HC-05 Bluetooth module
- L298N motor driver module
- 2 × BO DC gear motors with wheels
- 3 × Flame sensors
- 5V water pump
- SG90 servo motor
- Relay module
- LED indicator
- Buzzer
- 18650 lithium-ion batteries with holder
- DC-DC buck converter
- Wooden chassis base
- Nuts and bolts for assembly
- Jumper wires and breadboard

Fig. 2. *The Bluetooth controlled RC fire car look*

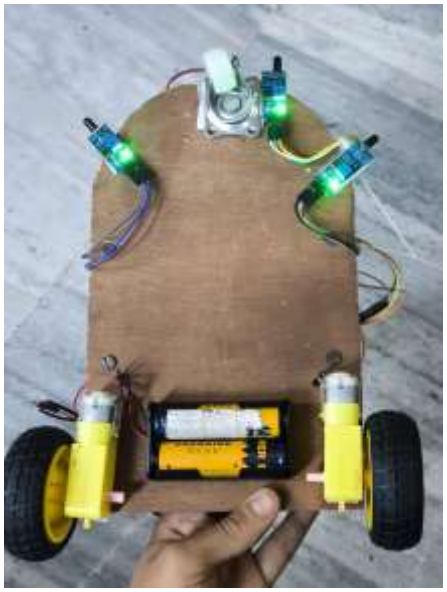

Fig. 3. *The Bluetooth controlled RC fire car back*

IV. Hardware Implementation and Assembly Procedure

Step 1: Identification of Components

All electronic and mechanical components required for the project were collected and verified before assembly. These included the Arduino Nano, HC-05 module, flame sensors, motor driver, water pump, servo motor, and supporting mechanical parts.

Step 2: Chassis Preparation

A **custom wooden base** was used as the robot chassis. The structure was designed as a **two-storey platform**, allowing electronic components to be mounted on the upper layer while the motors and batteries were placed on the lower layer.

This arrangement helped separate the **water tank from sensitive electronic components**, improving safety and organization.

Step 3: Motor Installation

The BO DC motors were mounted on the chassis using screws. The rubber wheels were attached securely to the motor shafts to ensure smooth movement of the robot.

Step 4: Sensor Installation

Three flame sensors were installed at the front of the robot to detect fire from different directions (left, center, and right). This configuration improves fire detection accuracy.

Step 5: Water Pump Installation

A small **5V water pump** was installed along with a water pipe and nozzle. The pump is controlled through a relay module connected to the Arduino.

Step 6: Servo Motor Installation

A **servo motor** was mounted near the water nozzle to control the direction of the water spray.

Step 7: Power Supply Installation

The system uses **18650 lithium-ion batteries** connected through a **DC-DC buck converter** to regulate voltage for the electronic components.

V. Working Principle

After powering the system, the **HC-05 Bluetooth module** pairs with a smartphone. The user sends control commands such as forward, backward, left, and right using a Bluetooth terminal application.

The Arduino Nano receives these commands and controls the motor driver to move the robot accordingly.

When a flame sensor detects fire:

1. The robot stops moving
2. LED and buzzer are activated
3. Servo motor aligns the water nozzle
4. The relay activates the water pump
5. Water is sprayed to extinguish the fire

VI. Software Implementation

The Arduino Nano is programmed using the **Arduino IDE**. The program reads serial data from the Bluetooth module and processes commands to control the motors.

Flame sensor inputs are monitored continuously. When fire is detected, the program activates the relay module and controls the servo motor for water direction.

VII. Challenges Faced and Lessons Learned

During the development of this project, several practical challenges were encountered. These challenges provided valuable learning experience in both hardware design and system integration.

1. Bluetooth Application Compatibility

One of the issues faced during the project was related to the **HC-05 Bluetooth module connectivity**. The module successfully connected with the **Serial Bluetooth Terminal application**, but it was not able to connect properly with several other Bluetooth controller applications.

To solve this problem, custom control buttons were created inside the **Serial Bluetooth Terminal app**, allowing the robot to be controlled by sending specific command characters such as **F (Forward), B (Backward), L (Left), R (Right), and S (Stop)**.

This experience helped in understanding the importance of **Bluetooth communication protocols and application compatibility**.

2. Water Tank Placement and Flow Issue

The robot was designed with a **two-layer wooden chassis structure**. This design was chosen to separate the **water tank from the electronic components**, which improves safety and protects sensitive electronics from water damage.

However, due to the **natural flow of water caused by gravity**, water sometimes continued to flow through the pipe even when the water pump was turned off. This happened because the water tank was positioned at a higher level than the pump and pipe outlet.

This issue highlighted the importance of **proper fluid flow control and component positioning in mechanical design**.

3. Water Spray Direction Control

Another challenge was maintaining a **straight and controlled water spray** from the pump nozzle. Since the pump and pipe system was newly implemented, it required several adjustments in the nozzle direction and servo motor alignment to achieve a stable water stream.

This process helped in understanding the **mechanical alignment and control of fluid systems**.

4. Component Mounting Stability

Initially, some components were mounted using **double-sided tape**. However, during robot movement this method proved unstable and components could shift or detach.

After observing this issue, it was realized that **hot glue (glue gun) or proper mounting brackets** would provide a stronger and more reliable attachment for the components.

This experience emphasized the importance of **proper mechanical mounting techniques in robotics projects**.

Lessons Learned

Through these challenges, several important lessons were learned:

- Proper testing of **Bluetooth communication apps** is necessary.
- **Mechanical design and fluid flow considerations** are important when working with water systems.
- Stable **component mounting methods** are essential for reliable robot operation.
- Practical project development often requires **iterative testing and improvements**.

VIII. Applications

• Educational robotics projects
• Fire detection demonstrations
• Safety and automation prototypes
• Remote controlled robotic systems

IX. Advantages and Limitations

Advantages

• Low cost implementation
• Wireless control using smartphone
• Compact design using Arduino Nano
• Effective fire detection system

Limitations

• Limited Bluetooth range
• Manual navigation required
• Small water capacity

X. Conclusion and Future Scope

The Bluetooth controlled fire fighting robot demonstrates an effective combination of **robotics, sensors, and wireless communication**. The system successfully detects fire and sprays water automatically while allowing manual control through a smartphone.

Future improvements may include **autonomous navigation, obstacle detection sensors, camera monitoring, and Wi-Fi based control systems**.

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